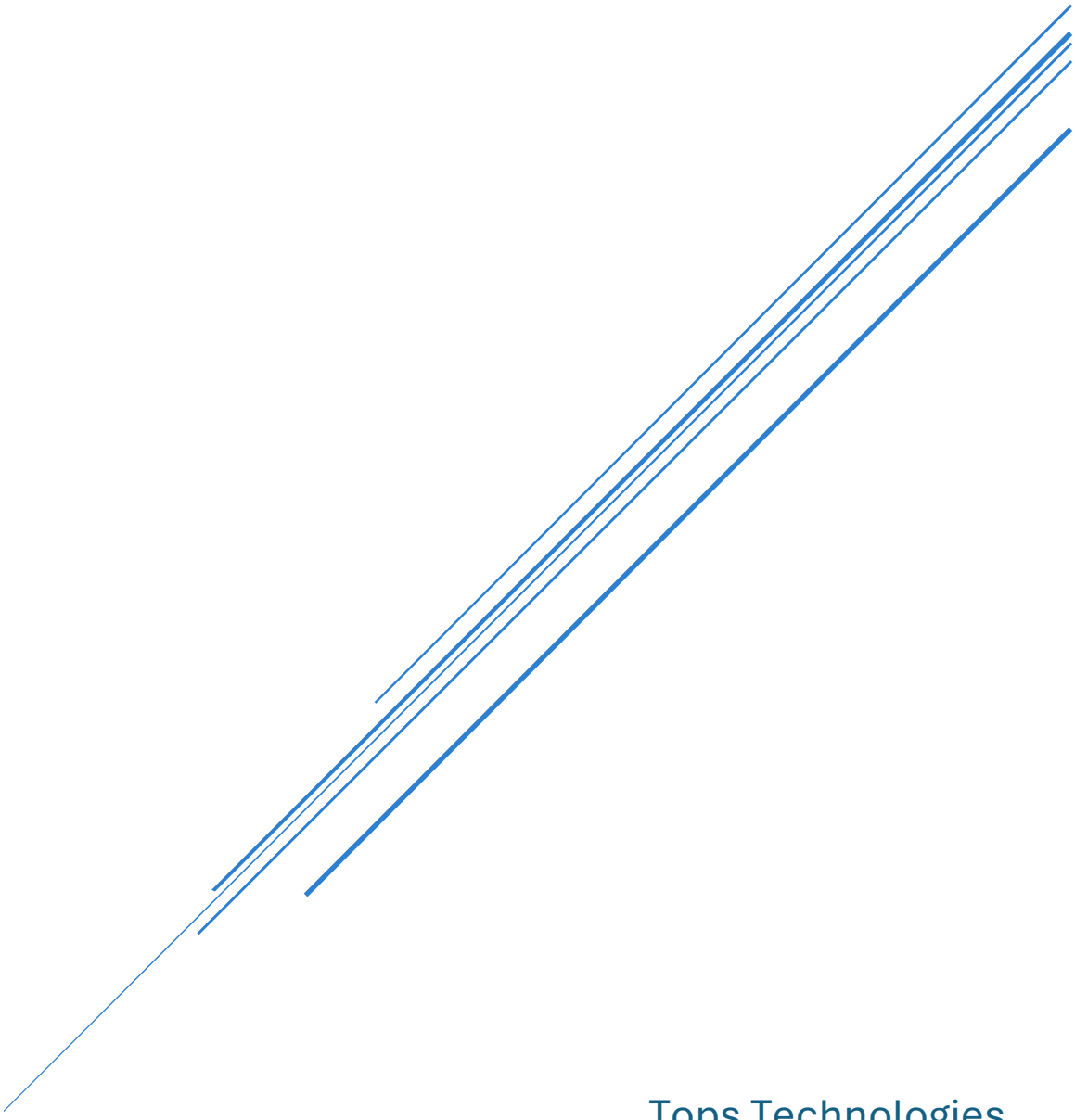
Introduction to Networks

Assignment - 1



Tops Technologies
Cyber Security

1. **Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, Wi-Fi router, ISP, and Instagram server. Label each part and the connections.**

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When I open Instagram on my phone, my phone connects to the Wi-Fi router using a wireless connection. The router sends my data to the Internet Service Provider (ISP), such as Jio or Airtel. The ISP connects my request to the internet, which carries the data to Instagram's servers. The Instagram server processes my request and sends back the content, such as photos, videos, and messages, to my phone.

2. **List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).**

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No.	Device	Type of Network	Explanation
1	Smartphone	WAN / WLAN	Connects to the internet using mobile data or Wi-Fi.
2	Laptop	WLAN / LAN	Connects to a Wi-Fi network or wired local network.
3	Wi-Fi Router	LAN	Connects multiple devices within a home or office network.
4	Smart TV	WLAN / LAN	Connects to the internet to stream videos and movies.
5	Smartwatch	PAN	Connects to a smartphone using Bluetooth.

Network types:

- **LAN: Local Area Network — connects devices in a small area, such as a home or office.**
- **WAN: Wide Area Network — connects networks over large geographical areas.**
- **WLAN: Wireless Local Area Network — connects devices wirelessly using Wi-Fi.**
- **PAN: Personal Area Network — connects personal devices over a short distance, usually using Bluetooth.**

3. **Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.**

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- **Internet**: The internet is a global network that connects millions of computers and devices worldwide. We can use it to access websites like Google, YouTube, and Instagram.

- **Intranet**: An intranet is a private network used within an organization, such as a college or company. It is accessible only to authorized users.

- **Example**: When I connect my phone to my college's public Wi-Fi and browse YouTube, I am using the internet. If my college has a private student portal accessible only to authorized students and staff, that portal is part of the college's intranet.

4. **Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.**

   Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

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- When I open WhatsApp on my smartphone, my device connects to the internet through Wi-Fi or mobile data.

- When I search for a message, WhatsApp processes the search request on my device.

- If the search requires server communication, the data travels through the router or mobile network, ISP, and internet to WhatsApp's server.

- The server processes the request and sends the required response back to my device.

- WhatsApp displays the search results on my phone.